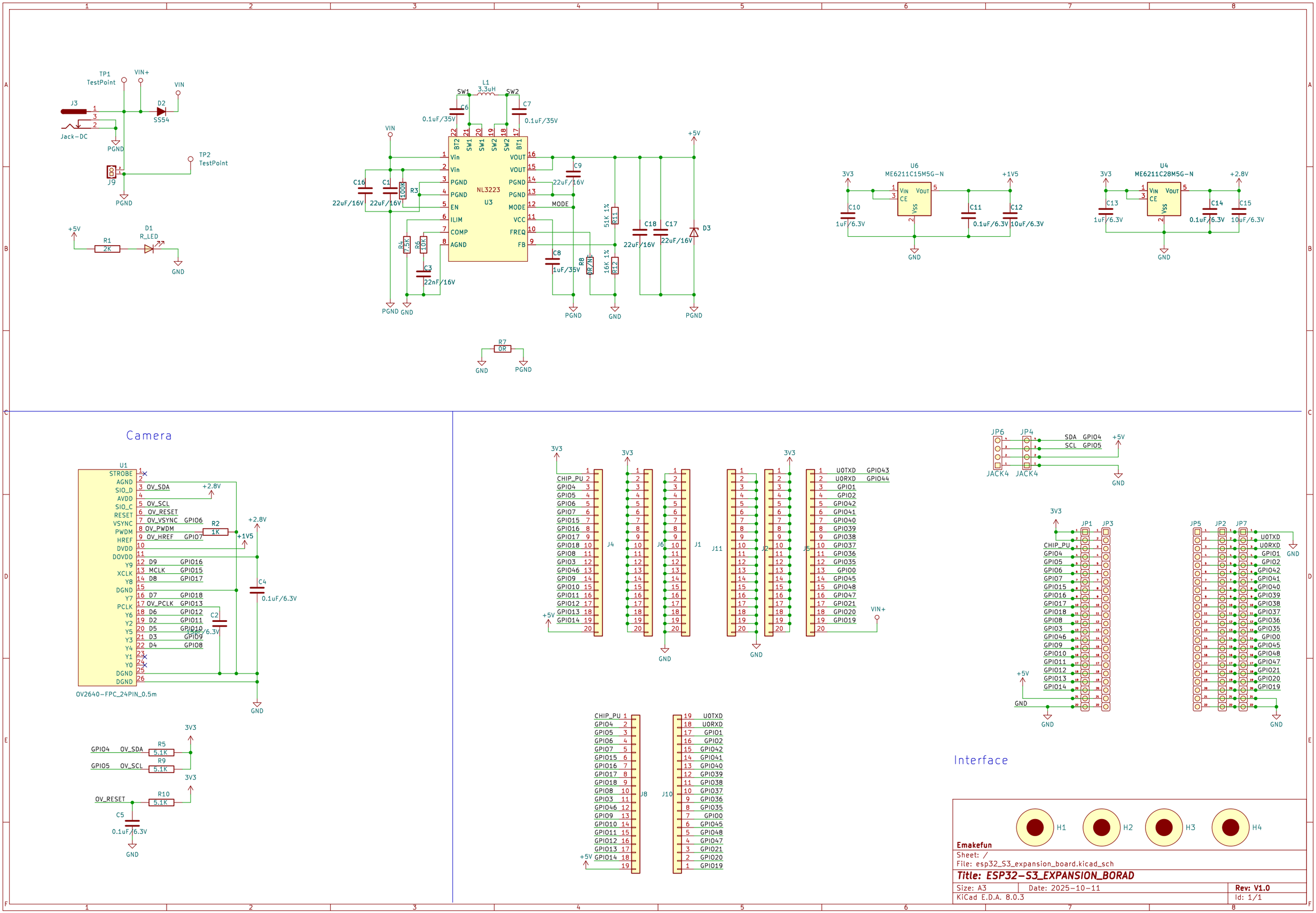
TP1
TestPoint
VIN+
VIN
J3
D2
SS54
Jack-DC
PGND
J9
TP2
TestPoint
+5V
R1
2K
D1
R_LED
GND
L1
3.3uH
SW1
SW2
C6
0.1uF/35V
C7
0.1uF/35V
NL3223
U3
BT2
SW1
SW2
BT1
Vin
PGND
EN
ILIM
COMP
AGND
VOUT
MODE
VCC
FREQ
FB
C16
22uF/16V
C1
22uF/16V
100K
R3
R4
7.5K
R6
10K
C3
22nF/16V
C9
22uF/16V
C8
1uF/35V
R8
DB/NC
51K 1%
R11
16K 1%
R12
C18
22uF/16V
C17
22uF/16V
D3
R7
0R
U6
ME6211C15M5G-N
3V3
C10
1uF/6.3V
VIN
CE
VOUT
VSS
C11
0.1uF/6.3V
+1V5
C12
10uF/6.3V
U4
ME6211C28M5G-N
C13
1uF/6.3V
C14
0.1uF/6.3V
+2.8V
C15
10uF/6.3V
Camera
U1
STROBE
AGND
SIO_D
AVDD
SIO_C
RESET
VSYNC
PWDM
HREF
DVDD
DOVDD
Y9
XCLK
Y8
DGND
Y7
PCLK
Y6
Y2
Y5
Y3
Y4
Y1
Y0
OV_SDA
OV_SCL
OV_RESET
OV_VSYNC
OV_PWDM
OV_HREF
D9
MCLK
D8
D7
OV_PCLK
D6
D2
D5
D3
D4
GPIO6
GPIO7
GPIO16
GPIO15
GPIO17
GPIO18
GPIO13
GPIO12
GPIO11
GPIO10
GPIO9
GPIO8
R2
1K
C4
0.1uF/6.3V
C2
0.1uF/6.3V
OV2640-FPC_24PIN_0.5m
R5
5.1K
R9
5.1K
R10
5.1K
C5
0.1uF/6.3V
Interface
JP6
JP4
JACK4
SDA GPIO4
SCL GPIO5
J4
J6
J1
J11
J2
J5
CHIP_PU
GPIO4
GPIO5
GPIO6
GPIO7
GPIO15
GPIO16
GPIO17
GPIO18
GPIO8
GPIO3
GPIO46
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
U0TXD GPIO43
U0RXD GPIO44
GPIO1
GPIO2
GPIO42
GPIO41
GPIO40
GPIO39
GPIO38
GPIO37
GPIO36
GPIO35
GPIO0
GPIO45
GPIO48
GPIO47
GPIO21
GPIO20
GPIO19
VIN+
J8
J10
JP1
JP3
JP5
JP2
JP7
H1
H2
H3
H4
Emakefun
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Title: ESP32-S3_EXPANSION_BORAD
Size: A3
Date: 2025-10-11
Rev: V1.0
KiCad E.D.A. 8.0.3
Id: 1/1